Supplement to

Microbial decomposition of kelp detritus in response to light and temperature

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Table S1. Bacterial genera that are confirmed alginate, laminarin, fucoidan, kelp or fucoid degraders. Up to four examples of the identified number of species are given.

Genus		Species	Studies
Vibrio	*natriegens, atlanticus, hyugaensis, rotiferianus*	28	15
Bacillus	*megaterium, licheniformis, zhangzhouensis, infantis*	25	13
Pseudoalteromonas	*agarivorans, distincta, prydzensis, aliena*	21	11
Alteromonas	*macleodii, carrageenovora, stellipolaris, addita*	9	11
Pseudomonas	*fluorescens, aeruginosa, grimontii, alginovora*	10	9
Zobellia	*russellii, galactanivorans, uliginosa*	3	8
Shewanella	*gaetbuli, algidipiscicola, pacifica, japonica*	7	6
Cellulophaga	*omnivescoria, lytica, fucicola, geojensis*	6	6
Flavobacterium	*algicola, faecale*	2	6
Halomonas	*marina, venusta, denitrificans*	3	5
Microbulbifer	*mangrovi, elongatus, celer*	3	5
Maribacter	*forsetii, stanieri*	2	5
Sphingomonas	*paucimobilis, leidyi*	2	4
Psychrobacter	*maritimus, nivimaris, cryohalolentis, alimentarius*	8	3
Cytophaga	*diffluens, uliginosa*	2	3
Photobacterium	*swingsii*	1	3
Cobetia	*litoralis*	1	3
Streptomyces		0	3
Colwellia		0	3
Erythrobacter	*longus*	1	2
Brevibacterium	*celere*	1	2
Gracilibacillus	*halotolerans*	1	2
Fucobacter	*marina*	1	2
Marinomonas	*ushuaiensis*	1	2
Algibacter	*lectus*	1	2
Marinobacter	*hydrocarbonoclasticus*	1	2
Tamlana	*agarivorans*	1	2
Agarivorans	*albus*	1	2
Enterobacter	*tabaci*	1	2
Alcaligenes		0	2
Glaciecola		0	2
Psychromonas		0	2
Polaribacter		0	2

Halobacillus	*dabanensis, trueperi, kuroshimensis, halophilus*	6	1
Paenibacillus	*jamilae, lautus, odorifer, taichungensis*	4	1
Fictibacillus	*nanhaiensis, phosphorivorans*	2	1
Lutibacter	*litoralis, crassostreae*	2	1
Algoriphagus	*winogradskyi, chordae*	2	1
Enterovibrio	*norvegicus, calviensis*	2	1
Paracoccus	*homiensis, fistulariae*	2	1
Pontiella	*desulfatans, sulfatireligans*	2	1
Loktanella	*agnita*	1	1
Tateyamaria	*omphalii/pelophila*	1	1
Phaeobacter	*arcticus*	1	1
Jannaschia	*donghaensis*	1	1
Labrenzia	*alba*	1	1
Altererythrobacter	*indicus*	1	1
Croceicoccus	*marinus*	1	1
Alginovibrio	*aquatilis*	1	1
Deleya	*marina*	1	1
Fucophilus	*fucoidanolyticus*	1	1
Mesonia	*algae*	1	1
Lentimonas	*marisflavi*	1	1
Marineflexile	*fucanivorans*	1	1
Limimaricola	*soesokkakensis*	1	1
Marinovum	*algicola*	1	1
Pseudophaeobacter	*arcticus*	1	1
Celeribacter	*halophilus*	1	1
Thalassobacillus	*devorans*	1	1
Lacinutrix	*gracilariae*	1	1
Winogradskyella	*sediminis*	1	1
Alcanivorax	*venustensis*	1	1
Leclercia	*adecarboxylata*	1	1
Planomicrobium	*okeanokoites*	1	1
Isoptericola	*halotolerans*	1	1
Lysinibacillus	*macroides*	1	1
Microbacterium	*oxydans*	1	1
Pseudooceanicola	*algae*	1	1
Hydrogenophaga	*taeniospralis*	1	1
Aquimarina	*latercula*	1	1
Huaishuia	*halophila*	1	1
Pseudozobellia	*thermophila*	1	1
Aliivibrio	*fischeri*	1	1
Clostridium	*butyricum*	1	1
Saccharophagus	*degradans*	1	1
Sinomicrobium	*oceani*	1	1
Rhodopirellula	*gimesia*	1	1
Neorhodopirellula	*lusitana*	1	1
Roseobacter		0	1
Alginomonas		0	1
Moraxella		0	1
Ochrobactrum		0	1
Arenibacter		0	1
Gramella		0	1
Micrococcus		0	1

Psychrobium	0	1
Wenyingzhuangia	0	1
Tenacibaculum	0	1
Leucothrix	0	1
Psychrosphaera	0	1
Formosa	0	1
Idiomarina	0	1
Litoreibacter	0	1
Paraglaciecola	0	1
Rheinheimera	0	1
Rhodococcus	0	1
Sulfitobacter	0	1
Salegentibacter	0	1
Prevotella	0	1
Staphylococcus	0	1
Simiduia	0	1

References. Studies identifying bacterial genera that degrade macroalgal polysaccharides.
1. (Bengtsson et al. 2011)
2. (Salaün et al. 2012)
3. (Bengtsson & Øvreås 2010)
4. (Ivanova et al. 2004)
5. (Goecke et al. 2010)
6. (Brunet et al. 2022)
7. (Thomas et al. 2021)
8. (Liang et al. 2021)
9. (Laycock 1974)
10. (Shiba & Taga 1980)
11. (Thomas et al. 2017)
12. (Sichert et al. 2020)
13. (Egan et al. 2014)
14. (Kumar et al. 2022)
15. (Martin et al. 2014)
16. (Xu et al. 2021)
17. (Cha et al. 2023)
18. (Wang et al. 2021)
19. (Barbato et al. 2022)
20. (Rodrigues et al. 2017)
21. (Sánchez Hinojosa et al. 2018)
22. (Alvarado & Leiva 2017)
23. (Wang et al. 2017)
24. (Naik et al. 2019)
25. (Jonnadula et al. 2009)
26. (Uchida 1996)
27. (Lin et al. 2018)
28. (Küpper et al. 2002)
29. (Erasmus et al. 1997)
30. (Kim et al. 2013)

31. (Liu et al. 2019)
32. (Wolter et al. 2021)
33. (Martin et al. 2016)
34. (Tang et al. 2011)
35. (Yamaguchi et al. 2019)
36. (Yang et al. 2021)
37. (Uchida et al. 1995)
38. (Nedashkovskaya et al. 2018)
39. (Nedashkovskaya et al. 2005)
40. (Nedashkovskaya et al. 2006)
41. (Goecke et al. 2013)
42. (Lavin et al. 2013)
43. (Ibrahim et al. 2015)
44. (Maruyama et al. 2019)
45. (Kouzuma & Fujii 2018)
46. (Corzett et al. 2018)
47. (Barbeyron et al. 2016)
48. (Mitulla et al. 2016)
49. (Neumann et al. 2015)
50. (Xia et al. 2023)
51. (Martin et al. 2015)
52. (Gobet et al. 2018)
53. (Koch et al. 2019a)
54. (Kalenborn et al. 2024)
55. (Johansen et al. 1999)
56. (Jain & Krishnan 2017)
57. (Kim et al. 2008)
58. (Nakamura et al. 2022)
59. (Liu et al. 2024a)
60. (Liu et al. 2024b)
61. (Zhu et al. 2018)
62. (Sun et al. 2022)
63. (Gong et al. 2024)
64. (Williams et al. 2013)
65. (Sakatoku et al. 2012)
66. (Wang et al. 2016)
67. (Nakata et al. 2024)
68. (Lavín et al. 2016)
69. (Shobharani et al. 2013)
70. (Valdehuesa et al. 2018)
71. (Mohapatra 2025)
72. (Sakai et al. 2002)
73. (Imran et al. 2017)
74. (Vashist et al. 2013)
75. (Wakabayashi et al. 2012)

76. (Zeb et al. 2024)
77. (Kang & Kim 2015)
78. (Wang et al. 2014)
79. (Gomare et al. 2011)
80. (Li et al. 2011)
81. (Tawara et al. 2015)
82. (Saravanan et al. 2024)
83. (Petchimuthu et al. 2024)
84. (Ramya et al. 2023)
85. (Jegatheesan et al. 2017)
86. (Mohapatra 2021)
87. (Hung et al. 2025)
88. (Koch et al. 2019b)
89. (Pérez-Cruz et al. 2024)
90. (Hettle et al. 2019)
91. (Zhu et al. 2016)
92. (Cheng et al. 2020)
93. (Chen et al. 2018)

Alvarado R, Leiva S (2017) Agar-degrading bacteria isolated from Antarctic macroalgae. *Folia Microbiol (Praha)* 62, 409–416. https://doi.org/10.1007/s12223-017-0511-1.

Barbato M, Vacchini V, Engelen AH, Patania G, Mapelli F, Borin S, Crotti E (2022) What lies on macroalgal surface: diversity of polysaccharide degraders in culturable epiphytic bacteria. *AMB Express* 12, 98. https://doi.org/10.1186/s13568-022-01440-8.

Barbeyron T, Thomas F, Barbe V, Teeling H, Schenowitz C, Dossat C, Goesmann A, Leblanc C, Oliver Glöckner F, Czjzek M et al. (2016) Habitat and taxon as driving forces of carbohydrate catabolism in marine heterotrophic bacteria: example of the model algae-associated bacterium Zobellia galactanivorans Dsij^{T}. *Environ Microbiol* 18, 4610–4627. https://doi.org/10.1111/1462-2920.13584.

Bengtsson MM, Øvreås L (2010) Planctomycetes dominate biofilms on surfaces of the kelp Laminaria hyperborea. *BMC Microbiol* 10, 261. https://doi.org/10.1186/1471-2180-10-261.

Bengtsson MM, Sjøtun K, Storesund JE, Øvreås J (2011) Utilization of kelp-derived carbon sources by kelp surface-associated bacteria. *Aquat Microb Ecol* 62, 191–199. 10.3354/ame01477.

Brunet M, Le Duff N, Barbeyron T, Thomas F (2022) Consuming fresh macroalgae induces specific catabolic pathways, stress reactions and Type IX secretion in marine flavobacterial pioneer degraders. *ISME J* 16, 2027–2039. 10.1038/s41396-022-01251-6.

Cha Q-Q, Liu S-S, Dang Y-R, Ren X-B, Xu F, Li P-Y, Chen X-L, Wang P, Zhang X-Y, Zhang Y-Z et al. (2023) Ecological function and interaction of different bacterial groups during alginate processing in coastal seawater community. *Environ Int* 182, 108325. https://doi.org/10.1016/j.envint.2023.108325.

Chen P, Zhu Y, Men Y, Zeng Y, Sun Y (2018) Purification and Characterization of a Novel Alginate Lyase from the Marine Bacterium Bacillus sp. Alg07. *Mar Drugs* 16, 86. https://pubmed.ncbi.nlm.nih.gov/29522433.

Cheng W, Yan X, Xiao J, Chen Y, Chen M, Jin J, Bai Y, Wang Q, Liao Z, Chen Q (2020) Isolation, identification, and whole genome sequence analysis of the alginate-degrading bacterium Cobetia sp. cqz5-12. *Sci Rep* 10, 10920. https://pubmed.ncbi.nlm.nih.gov/32616809.

Corzett CH, Elsherbini J, Chien DM, Hehemann J-H, Henschel A, Preheim SP, Yu X, Alm EJ, Polz MF (2018) Evolution of a vegetarian *Vibrio*: metabolic specialization of *Vibrio breoganii* to macroalgal substrates. *Journal of Bacteriology* 200, 10.1128/jb. 00020–18. https://doi.org/10.1128/jb.00020-18.

Egan S, Fernandes ND, Kumar V, Gardiner M, Thomas T (2014) Bacterial pathogens, virulence mechanism and host defence in marine macroalgae. *Environ Microbiol* 16, 925–938. https://doi.org/10.1111/1462-2920.12288.

Erasmus JH, Cook PA, Coyne VE (1997) The role of bacteria in the digestion of seaweed by the abalone Haliotis midae. *Aquaculture* 155, 377–386. https://doi.org/10.1016/S0044-8486(97)00112-9.

Gobet A, Barbeyron T, Matard-Mann M, Magdelenat G, Vallenet D, Duchaud E, Michel G (2018) Evolutionary Evidence of Algal Polysaccharide Degradation Acquisition by Pseudoalteromonas carrageenovora 9^{T} to Adapt to Macroalgal Niches. *Front Microbiol* 9, 2740. https://pubmed.ncbi.nlm.nih.gov/30524390.

Goecke F, Labes A, Wiese J, Imhoff JF (2010) Chemical interactions between marine macroalgae and bacteria. *Marine ecology progress series* 409, 267–299. https://doi.org/10.3354/meps08607.

Goecke F, Thiel V, Wiese J, Labes A, Imhoff JF (2013) Algae as an important environment for bacteria – phylogenetic relationships among new bacterial species isolated from algae. *Phycologia* 52, 14–24. https://doi.org/10.2216/12-24.1.

Gomare S, Kim HA, Ha JH, Lee MW, Park JM (2011) Isolation of the polysaccharidase-producing bacteria from the gut of sea snail, Batillus cornutus. *Korean Journal of Chemical Engineering* 28, 1252–1259. https://doi.org/10.1007/s11814-010-0506-y.

Gong Y, Shang D-D, Sun C-L, Du Z-J, Chen G-J (2024) Direct Degradation of Fresh and Dried Macroalgae by Agarivorans albus B2Z047. *Mar Drugs* 22, 203. https://pubmed.ncbi.nlm.nih.gov/38786594.

Hettle AG, Hobbs JK, Pluvinage B, Vickers C, Abe KT, Salama-Alber O, McGuire BE, Hehemann J-H, Hui JPM, Berrue F et al. (2019) Insights into the □/□-carrageenan metabolism pathway of some marine Pseudoalteromonas species. *Commun Biol* 2, 474. https://pubmed.ncbi.nlm.nih.gov/31886414.

Hung Y-HR, Chang Y-H, Lin H-J, Chiang L-H, Lin H-TV (2025) Enhanced antioxidant activity of brown seaweed Laminaria japonica by fermentation using isolated Bacillus subtilis. *Bioresour Bioprocess* 12, 69. https://pubmed.ncbi.nlm.nih.gov/40613959.

Ibrahim HAH, Beltagy EA, El-Din NGS, El Zokm GM, El-Sikaily AM, Abu-Elela GM (2015) Seaweeds agarophytes and associated epiphytic bacteria along Alexandria coastline, Egypt, with emphasis on the evaluation and extraction of agar and agarose. *Revista de biología marina y oceanografía* 50, 545–561. https://doi.org/10.4067/S0718-19572015000400012.

Imran M, Pant P, Shanbhag YP, Sawant SV, Ghadi SC (2017) Genome Sequence of Microbulbifer mangrovi DD-13^{T} Reveals Its Versatility to Degrade Multiple Polysaccharides. *Mar Biotechnol (NY)* 19, 116–124. https://pubmed.ncbi.nlm.nih.gov/28161851.

Ivanova EP, Christen R, Alexeeva YV, Zhukova NV, Gorshkova NM, Lysenko AM, Mikhailov VV, Nicolau DV (2004) Brevibacterium celere sp. nov., isolated from degraded thallus of a brown alga. *Int J Syst Evol Microbiol* 54, 2107–2111. https://pubmed.ncbi.nlm.nih.gov/15545442.

Jain A, Krishnan KP (2017) A glimpse of the diversity of complex polysaccharide-degrading culturable bacteria from Kongsfjorden, Arctic Ocean. *Annals of Microbiology* 67, 203–214. https://doi.org/10.1007/s13213-016-1252-0.

Jegatheesan A, Sudhakar MP, Poonam C, Perumal K, Arunkumar K (2017) Isolation and characterization of alginate-degrading bacteria Sinomicrobium oceani. *Biomass Conversion and Biorefinery* 7, 51–58. https://doi.org/10.1007/s13399-016-0212-z.

Johansen JE, Nielsen P, Sjøholm C (1999) Description of Cellulophaga baltica gen. nov., sp. nov. and Cellulophaga fucicola gen. nov., sp. nov. and reclassification of [Cytophaga] lytica to Cellulophaga lytica gen. nov., comb. nov. *International Journal of Systematic and Evolutionary Microbiology* 49, 1231–1240. https://doi.org/10.1099/00207713-49-3-1231.

Jonnadula R, Verma P, Shouche YS, Ghadi SC (2009) Characterization of Microbulbifer strain CMC-5, a new biochemical variant of Microbulbifer elongatus type strain DSM6810T isolated from decomposing seaweeds. *Current microbiology* 59, 600–607. https://doi.org/10.1007/s00284-009-9480-1.

Kalenborn S, Zühlke D, Reintjes G, Riedel K, Amann RI, Harder J (2024) Genes for laminarin degradation are dispersed in the genomes of particle-associated Maribacter species. *Front Microbiol* 15, 1393588. https://pubmed.ncbi.nlm.nih.gov/39188312.

Kang S, Kim JK (2015) Reuse of red seaweed waste by a novel bacterium, Bacillus sp. SYR4 isolated from a sandbar. *World J Microbiol Biotechnol* 31, 209–217. https://pubmed.ncbi.nlm.nih.gov/25407816.

Kim EJ, Fathoni A, Jeong G-T, Jeong HD, Nam T-J, Kong I-S, Kim JK (2013) Microbacterium oxydans, a novel alginate- and laminarin-degrading bacterium for the reutilization of brown-seaweed waste. *J Environ Manage* 130, 153–159. https://doi.org/10.1016/j.jenvman.2013.08.064.

Kim W-J, Kim S-M, Lee Y-H, Kim HG, Kim H-K, Moon SH, Suh H-H, Jang K-H, Park Y-I (2008) Isolation and characterization of marine bacterial strain degrading fucoidan from Korean Undaria pinnatifida Sporophylls. *Journal of microbiology and biotechnology* 18, 616–623. https://europepmc.org/article/med/18467852.

Koch H, Freese HM, Hahnke RL, Simon M, Wietz M (2019a) Adaptations of Alteromonas sp. 76-1 to Polysaccharide Degradation: A CAZyme Plasmid for Ulvan Degradation and Two Alginolytic Systems. *Front Microbiol* 10, 504. https://pubmed.ncbi.nlm.nih.gov/30936857.

Koch H, Dürwald A, Schweder T, Noriega-Ortega B, Vidal-Melgosa S, Hehemann J-H, Dittmar T, Freese HM, Becher D, Simon M et al. (2019b) Biphasic cellular adaptations and ecological implications of Alteromonas macleodii degrading a mixture of algal polysaccharides. *ISME J* 13, 92–103. https://pubmed.ncbi.nlm.nih.gov/30116038.

Kouzuma S, Fujii K (2018) Biochemical characteristics of cellulose and a green alga degradation by Gilvimarinus japonicas 12-2^{T}, and its application potential for seaweed saccharification. *Biosci Biotechnol Biochem* 82, 2198–2204. https://doi.org/10.1080/09168451.2018.1516542.

Kumar P, Verma A, Sundharam SS, Ojha AK, Krishnamurthi S (2022) Exploring Diversity and Polymer Degrading Potential of Epiphytic Bacteria Isolated from Marine Macroalgae. *Microorganisms* 10, 2513. https://doi.org/10.3390/microorganisms10122513.

Küpper FC, Müller DG, Peters AF, Kloareg B, Potin P (2002) Oligoalginate recognition and oxidative burst play a key role in natural and induced resistance of sporophytes of Laminariales. *Journal of Chemical Ecology* 28, 2057–2081. 10.1023/A:1020706129624.

Lavin P, Gallardo-Cerda J, Torres-Diaz C, Asencio G, Gonzalez M (2013) Cepa antártica de Bacillus sp., con actividad extracelular de tipo agarolítica y alginatoliasa. *Gayana (concepción)* 77, 75–82. https://doi.org/10.4067/S0717-65382013000200001.

Lavín P, Atala C, Gallardo-Cerda J, Gonzalez-Aravena M, De La Iglesia R, Oses R, Torres-Díaz C, Trefault N, Molina-Montenegro MA, Laughinghouse IV HD (2016) Isolation and characterization of an Antarctic Flavobacterium strain with agarase and alginate lyase activities. *Polish Polar Research* 37, 403–419. https://doi.org/10.1515/popore-2016-0021.

Laycock RA (1974) The detrital food chain based on seaweeds. I. Bacteria associated with the surface of Laminaria fronds. *Marine Biology* 25, 223–231. https://doi.org/10.1007/bf00394968.

Li L, Jiang X, Guan H, Wang P, Guo H (2011) Three alginate lyases from marine bacterium Pseudomonas fluorescens HZJ216: purification and characterization. *Appl Biochem Biotechnol* 164, 305–317. https://pubmed.ncbi.nlm.nih.gov/21170604.

Liang J, Liu J, Zhan Y, Zhou S, Xue CX, Sun C, Lin Y, Luo C, Wang X, Zhang XH (2021) Succession of marine bacteria in response to Ulva prolifera-derived dissolved organic matter. *Environ Int* 155, 106687. https://doi.org/10.1016/j.envint.2021.106687.

Lin JD, Lemay MA, Parfrey LW (2018) Diverse bacteria utilize alginate within the microbiome of the giant kelp *Macrocystis pyrifera*. *Front Microbiol* 9, 1914. 10.3389/fmicb.2018.01914.

Liu G-L, Wu S-L, Sun Z, Xing M-D, Chi Z-M, Liu Y-J (2024a) □-Carrageenan catabolism is initiated by key sulfatases in the marine bacterium Pseudoalteromonas haloplanktis LL1. *Appl Environ Microbiol* 90, e0025524. https://doi.org/10.1128/aem.00255-24.

Liu N, Kivenson V, Peng X, Cui Z, Lankiewicz TS, Gosselin KM, English CJ, Blair EM, O'Malley MA, Valentine DL (2024b) Pontiella agarivorans sp. nov., a novel marine anaerobic bacterium capable of degrading macroalgal polysaccharides and fixing nitrogen. *Appl Environ Microbiol* 90, e0091423. https://doi.org/10.1128/aem.00914-23.

Liu X, Chen Y, Zhong M, Chen W, Lin Q, Du H (2019) Isolation and pathogenicity identification of bacterial pathogens in bleached disease and their physiological effects on the red macroalga Gracilaria lemaneiformis. *Aquatic Botany* 153, 1–7. https://doi.org/10.1016/j.aquabot.2018.11.002.

Martin M, Portetelle D, Michel G, Vandenbol M (2014) Microorganisms living on macroalgae: diversity, interactions, and biotechnological applications. *Appl Microbiol Biotechnol* 98, 2917–2935. https://doi.org/10.1007/s00253-014-5557-2.

Martin M, Barbeyron T, Martin R, Portetelle D, Michel G, Vandenbol M (2015) The Cultivable Surface Microbiota of the Brown Alga Ascophyllum nodosum is Enriched in Macroalgal-Polysaccharide-Degrading Bacteria. *Front Microbiol* 6, 1487. https://pubmed.ncbi.nlm.nih.gov/26734000.

Martin M, Vandermies M, Joyeux C, Martin R, Barbeyron T, Michel G, Vandenbol M (2016) Discovering novel enzymes by functional screening of plurigenomic libraries from alga-

associated Flavobacteriia and Gammaproteobacteria. *Microbiol Res* 186-187, 52–61. https://doi.org/10.1016/j.micres.2016.03.005.
Maruyama Y, Hashimoto W, Murata K (2019) Structural studies on bacterial system used in the recognition and uptake of the macromolecule alginate. *Biosci Biotechnol Biochem* 83, 794–802. https://doi.org/10.1080/09168451.2019.1578642.
Mitulla M, Dinasquet J, Guillemette R, Simon M, Azam F, Wietz M (2016) Response of bacterial communities from California coastal waters to alginate particles and an alginolytic Alteromonas macleodii strain. *Environ Microbiol* 18, 4369–4377. https://doi.org/10.1111/1462-2920.13314.
Mohapatra BR (2025) Chryseobacterium barbadense sp. nov. Isolated from Pelagic Sargassum Seaweed Waste. *Curr Microbiol* 83, 111. https://pubmed.ncbi.nlm.nih.gov/41460337.
Mohapatra BR (2021) Comparative kinetic analysis of alginate lyase and mannanase co-produced via solid-state fermentation of cow dung supplemented with seaweed wastes by a novel Streptomyces sp. Alg-S23. *Biomass Conversion and Biorefinery* 11, 2013–2025. https://doi.org/10.1007/s13399-020-00658-5.
Naik MM, Naik D, Charya L, Mujawar SY, Vaingankar DC (2019) Application of marine bacteria associated with seaweed, Ulva lactuca, for degradation of algal waste. *Proceedings of the National Academy of Sciences, India Section B: Biological Sciences* 89, 1153–1160. https://doi.org/10.1007/s40011-018-1034-5.
Nakamura S, Yumioka J, Kachi S, Baba Y, Kawai S (2022) Bacterial and fungal gut microbiota of supralittoral talitrid amphipods feeding on brown macroalgae and paper. *PLoS One* 17, e0279834. https://journals.plos.org/plosone/article/file?id=10.1371/journal.pone.0279834&type=printable.
Nakata S, Takase R, Kawai S, Ogura K, Hashimoto W (2024) Potential role of alginate in marine bacteria-yeast interactions. *Appl Environ Microbiol* 90, e0168324. https://doi.org/10.1128/aem.01683-24.
Nedashkovskaya OI, Kim SB, Lysenko AM, Frolova GM, Mikhailov VV, Lee KH, Bae KS (2005) Description of Aquimarina muelleri gen. nov., sp. nov., and proposal of the reclassification of [Cytophaga] latercula Lewin 1969 as Stanierella latercula gen. nov., comb. nov. *Int J Syst Evol Microbiol 55*, 225–229. https://doi.org/10.1099/ijs.0.63349-0.
Nedashkovskaya OI, Vancanneyt M, Christiaens L, Kalinovskaya NI, Mikhailov VV, Swings J (2006) Aquimarina intermedia sp. nov., reclassification of Stanierella latercula (Lewin 1969) as Aquimarina latercula comb. nov. and Gaetbulimicrobium brevivitae Yoon et al. 2006 as Aquimarina brevivitae comb. nov. and emended description of the genus Aquimarina. *Int J Syst Evol Microbiol 56*, 2037–2041. https://doi.org/10.1099/ijs.0.64155-0.
Nedashkovskaya OI, Kim S-G, Stenkova AM, Kukhlevskiy AD, Zhukova NV, Mikhailov VV (2018) Aquimarina algiphila sp. nov., a chitin degrading bacterium isolated from the red alga Tichocarpus crinitus. *Int J Syst Evol Microbiol* 68, 892–898. https://doi.org/10.1099/ijsem.0.002606.
Neumann AM, Balmonte JP, Berger M, Giebel H-A, Arnosti C, Voget S, Simon M, Brinkhoff T, Wietz M (2015) Different utilization of alginate and other algal polysaccharides by marine Alteromonas macleodii ecotypes. *Environ Microbiol* 17, 3857–3868. https://doi.org/10.1111/1462-2920.12862.

Pérez-Cruz C, Moraleda-Montoya A, Liébana R, Terrones O, Arrizabalaga U, García-Alija M, Lorizate M, Martínez Gascueña A, García-Álvarez I, Nieto-Garai JA et al. (2024) Mechanisms of recalcitrant fucoidan breakdown in marine Planctomycetota. *Nat Commun* 15, 10906. https://pubmed.ncbi.nlm.nih.gov/39738071.

Petchimuthu R, Venkatesh S, Kannan S, Balakrishnan V (2024) Solid-state fermentation of brown seaweeds for the production of alginate lyase using marine bacterium Enterobacter tabaci RAU2C. *Folia Microbiol (Praha)* 69, 1083–1093. https://pubmed.ncbi.nlm.nih.gov/38401040.

Ramya P, Selvaraj K, Suthendran K, Sundar K, Vanavil B (2023) Optimization of alginate lyase production using Enterobacter tabaci RAU2C isolated from marine environment by RSM and ANFIS modelling. *Aquaculture International* 31, 3207–3237. https://doi.org/10.1007/s10499-023-01302-5.

Rodrigues V, Onime L, Huws S, Odaneth AA, Lali AM (2017) Diversity of ulvan and cellulose depolymerizing bacteria associated with the green macroalgae Ulva spp. *Journal of Applied Biotechnology & Bioengineering* 2, 136–142. https://doi.org/10.15406/jabb.2017.02.00037.

Sakai T, Kimura H, Kato I (2002) A marine strain of flavobacteriaceae utilizes brown seaweed fucoidan. *Mar Biotechnol (NY)* 4, 399–405. https://pubmed.ncbi.nlm.nih.gov/14961251.

Sakatoku A, Wakabayashi M, Tanaka Y, Tanaka D, Nakamura S (2012) Isolation of a novel Saccharophagus species (Myt-1) capable of degrading a variety of seaweeds and polysaccharides. *Microbiologyopen* 1, 2–12. https://pubmed.ncbi.nlm.nih.gov/22950007.

Salaün S, La Barre S, Dos Santos-Goncalvez M, Potin P, Haras D, Bazire A (2012) Influence of exudates of the kelp Laminaria digitata on biofilm formation of associated and exogenous bacterial epiphytes. *Microb Ecol* 64, 359–369. https://doi.org/10.1007/s00248-012-0048-4.

Sánchez Hinojosa V, Asenjo J, Leiva S (2018) Agarolytic culturable bacteria associated with three antarctic subtidal macroalgae. *World J Microbiol Biotechnol* 34, 73. https://doi.org/10.1007/s11274-018-2456-1.

Saravanan P, Chatterjee A, Kiran KJ, Bhowmick GD, Sappati PK, Nagarajan V (2024) Exploring Seaweed-Associated Marine Microbes: Growth Impacts and Enzymatic Potential for Sustainable Resource Utilization. *Indian J Microbiol* 64, 593–602. https://pubmed.ncbi.nlm.nih.gov/39011007.

Shiba T, Taga N (1980) Heterotrophic bacteria attached to seaweeds. *Journal of Experimental Marine Biology and Ecology* 47, 251–258. https://doi.org/10.1016/0022-0981(80)90042-8.

Shobharani P, Yogesh D, Halami PM, Sachindra NM (2013) Potential of Cellulase FromBacillus megateriumfor Hydrolysis ofSargassum. *Journal of Aquatic Food Product Technology* 22, 520–535. https://doi.org/10.1080/10498850.2012.670836.

Sichert A, Corzett CH, Schechter MS, Unfried F, Markert S, Becher D, Fernandez-Guerra A, Liebeke M, Schweder T, Polz MF et al. (2020) Verrucomicrobia use hundreds of enzymes to digest the algal polysaccharide fucoidan. *Nat Microbiol* 5, 1026–1039. https://doi.org/10.1038/s41564-020-0720-2.

Sun X-K, Gong Y, Shang D-D, Liu B-T, Du Z-J, Chen G-J (2022) Degradation of Alginate by a Newly Isolated Marine Bacterium Agarivorans sp. B2Z047. *Mar Drugs* 20, 254. https://pubmed.ncbi.nlm.nih.gov/35447927.

Tang J, Wang M, Zhou Q, Nagata S (2011) Improved composting of Undaria pinnatifida seaweed by inoculation with Halomonas and Gracilibacillus sp. isolated from marine environments. *Bioresour Technol* 102, 2925–2930. https://doi.org/10.1016/j.biortech.2010.11.064.

Tawara M, Sakatoku A, Tiodjio RE, Tanaka D, Nakamura S (2015) Cloning and Characterization of a Novel Agarase from a Newly Isolated Bacterium Simiduia sp. Strain TM-2 Able to Degrade Various Seaweeds. *Appl Biochem Biotechnol* 177, 610–623. https://pubmed.ncbi.nlm.nih.gov/26198024.

Thomas F, Bordron P, Eveillard D, Michel G (2017) Gene Expression Analysis of Zobellia galactanivorans during the Degradation of Algal Polysaccharides Reveals both Substrate-Specific and Shared Transcriptome-Wide Responses. *Front Microbiol* 8, 1808. https://doi.org/10.3389/fmicb.2017.01808.

Thomas F, Le Duff N, Wu TD, Cébron A, Uroz S, Riera P, Leroux C, Tanguy G, Legeay E, Guerquin-Kern JL (2021) Isotopic tracing reveals single-cell assimilation of a macroalgal polysaccharide by a few marine Flavobacteria and Gammaproteobacteria. *ISME J* 15, 3062–3075. https://doi.org/10.1038/s41396-021-00987-x.

Uchida M, Nakayama A, Abe S (1995) Distribution and characterization of bacteria capable of decomposing brown algae fronds in waters associated with *Laminaria* vegetation. *Fisheries science* 61, 117–120. https://doi.org/10.2331/fishsci.61.117.

Uchida M (1996) Formation of single cell detritus densely covered with bacteria during experimental degradation of *Laminaria japonica* thalli. *Fisheries Science* 62, 731–736. https://doi.org/10.2331/fishsci.62.731.

Valdehuesa KNG, Ramos KRM, Moron LS, Lee I, Nisola GM, Lee W-K, Chung W-J (2018) Draft Genome Sequence of Newly Isolated Agarolytic Bacteria Cellulophaga omnivescoria sp. nov. W5C Carrying Several Gene Loci for Marine Polysaccharide Degradation. *Curr Microbiol* 75, 925–933. https://doi.org/10.1007/s00284-018-1467-3.

Vashist P, Nogi Y, Ghadi SC, Verma P, Shouche YS (2013) Microbulbifer mangrovi sp. nov., a polysaccharide-degrading bacterium isolated from an Indian mangrove. *Int J Syst Evol Microbiol* 63, 2532–2537. https://pubmed.ncbi.nlm.nih.gov/23243090.

Wakabayashi M, Sakatoku A, Noda F, Noda M, Tanaka D, Nakamura S (2012) Isolation and characterization of Microbulbifer species 6532A degrading seaweed thalli to single cell detritus particles. *Biodegradation* 23, 93–105. https://pubmed.ncbi.nlm.nih.gov/21681519.

Wang D, Wang J, Zeng R, Wu J, Michael SV, Qu W (2021) The degradation activities for three seaweed polysaccharides of Shewanella sp. WPAGA9 isolated from deep-sea sediments. *J Basic Microbiol* 61, 406–418. https://doi.org/10.1002/jobm.202000728.

Wang M, Chen L, Liu Z, Zhang Z, Qin S, Yan P (2016) Isolation of a novel alginate lyase-producing Bacillus litoralis strain and its potential to ferment Sargassum horneri for biofertilizer. *Microbiologyopen* 5, 1038–1049. https://pubmed.ncbi.nlm.nih.gov/27440453.

Wang M, Chen L, Zhang Z, Wang X, Qin S, Yan P (2017) Screening of alginate lyase-excreting microorganisms from the surface of brown algae. *AMB Express* 7, 74. https://doi.org/10.1186/s13568-017-0361-x.

Wang Y, Li B, Zhao X, Piao M (2014) Isolation and characterization of a fucoidan-degrading bacterium from Laminaria japonica. *Journal of Ocean University of China* 13, 153–156. https://doi.org/10.1007/s11802-014-2011-0.

Williams AG, Withers S, Sutherland AD (2013) The potential of bacteria isolated from ruminal contents of seaweed-eating North Ronaldsay sheep to hydrolyse seaweed components and produce methane by anaerobic digestion in vitro. *Microb Biotechnol* 6, 45–52. https://doi.org/10.1111/1751-7915.12000.

Wolter LA, Wietz M, Ziesche L, Breider S, Leinberger J, Poehlein A, Daniel R, Schulz S, Brinkhoff T (2021) Pseudooceanicola algae sp. nov., isolated from the marine macroalga Fucus spiralis, shows genomic and physiological adaptations for an algae-associated lifestyle. *Syst Appl Microbiol* 44, 126166. https://doi.org/10.1016/j.syapm.2020.126166.

Xia H-F, Jia X-Y, Zhou Y-X, Du Z-J, Mu D-S, Chen G-J (2023) Comparative genomics reveal distinct potential of Tamlana sp. S12 for algal polysaccharide degradation. *Frontiers in Marine Science* 10, 7. https://doi.org/10.3389/fmars.2023.985514.

Xu F, Cha Q-Q, Zhang Y-Z, Chen X-L (2021) Degradation and Utilization of Alginate by Marine Pseudoalteromonas: a Review. *Appl Environ Microbiol* 87, e0036821. https://doi.org/10.1128/aem.00368-21.

Yamaguchi T, Narsico J, Kobayashi T, Inoue A, Ojima T (2019) Production of poly(3-hydroyxybutylate) by a novel alginolytic bacterium Hydrogenophaga sp. strain UMI-18 using alginate as a sole carbon source. *J Biosci Bioeng* 128, 203–208. https://doi.org/10.1016/j.jbiosc.2019.02.008.

Yang J, Cui D, Shen H (2021) Microbulbifer sp. SH-1 degrades brown seaweed via the synergistic effect of alginate lyase AlgSH6, AlgSH7 and AlgSH17. *Chinese Science Bulletin* 66, 4284–4294. https://doi.org/10.1360/tb-2021-0558.

Zeb A, Khan Y, He H, Zhang D, Shen S (2024) Molecular identification of Halomonas AZ07 and its multifunctional enzymatic activities to degrade Pyropia yezoensis under high-temperature condition. *Mol Biol Rep* 51, 816. https://pubmed.ncbi.nlm.nih.gov/39012539.

Zhu X, Li X, Shi H, Zhou J, Tan Z, Yuan M, Yao P, Liu X (2018) Characterization of a Novel Alginate Lyase from Marine Bacterium Vibrio furnissii H1. *Mar Drugs* 16, 30. https://pubmed.ncbi.nlm.nih.gov/29342949.

Zhu Y, Chen P, Bao Y, Men Y, Zeng Y, Yang J, Sun J, Sun Y (2016) Complete genome sequence and transcriptomic analysis of a novel marine strain Bacillus weihaiensis reveals the mechanism of brown algae degradation. *Sci Rep* 6, 38248. https://www.ncbi.nlm.nih.gov/pubmed/27901120.